

TechLifeProbe: A Technology Prototype to Probe How Data Sharing With Adolescents' Parents Improves Their Technology Abuse Through Mobile Phones

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ABSTRACT

This paper proposes TechLifeProbe to improve adolescents' technology abuse through sharing data with their parents. We hope to use TechLifeProbe to explore how data sharing among adolescents and parents helps them improve the technology abuse.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing**.

KEYWORDS

Technology abuse; Self-monitoring; Family support

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1 INTRODUCTION

Since screen use became a necessary part of our lives, technology abuse among adolescents has been the center of discussion. It refers to the problematic use of technology devices with negative impacts on individuals' well-being. Researchers have reported that many adolescents engage in maladaptive internet usage to deal with psychological difficulties [8] resulting from problematic relationships

with parents and/or peers [7]. What's worse, they often have different perception of technology abuse and reasoning of screen time data from parents [6]. As a result, they would constantly fight over this problem, to the degree that they have to seek medical support.

Existing technology solutions mostly rely on either self-control (e.g., Forest) [5]) or parental regulation (e.g., Google Family Link) [3]) on the screen usage. However, it appears that neither approach is effective in promoting productive discussion and reducing technology abuse [6]. For psychiatrists and parents, they require an easy-to-use tool to provide adolescents' screen time data required for meaningful analysis. For adolescents, such a system should also make it easier to self-monitor their screen use behavior and share their progress with their parents. To explore how to assist adolescents with technology abuse, further studies are required to understand the multi-faceted interaction with support forces from different stakeholders (e.g., their parents). Therefore, this study developed a phone-based support system (TechLifeProbe) to be employed in the technology probe study to collect users' experiences.

2 SYSTEM DESIGN AND PROTOTYPE

We propose the technology probe prototype based on our previous needs assessment study [6]. Both adolescents and parents carry their phones installed with an TechLifeProbe phone app. The adolescents wear wristband trackers (i.e., Fitbit Charge 2 [4]) to record their sleep hours and step counts. Moreover, the adolescents (or parents) can self-report (or assess) adolescents' technology use and health/well-being status once a day in a diary in order to keep track of the contextual feedback from both adolescents and parents. Finally, all collected data will be uploaded to the backend server. We will describe how we design and implement these component as follows.

2.1 TechLifeProbe Phone App

2.1.1 Main page. Figure 1(a) shows a screenshot of the main page, which visualizes the lifestyle factors, app use logs, and diary responses. The top of the main page visualizes logs collected from mobile phones and wristbands, and the bottom lists responses in the diary with a side-by-side comparison view of patients' self reports

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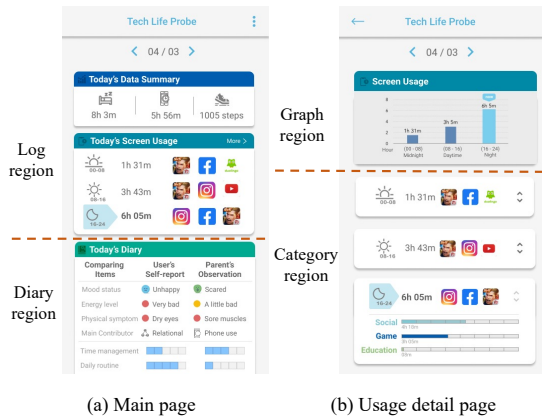


Figure 1: The screenshots of the TechLifeProbe phone app.

and parents' observations. The log region visualizes adolescents' device usage history and statistic automatically accessing by using built-in screen time APIs and and lifestyles recorded by using wristband trackers worn on the adolescents, respectively. There are one dashboard card and one screen usage card. The dashboard card shows the sleep hours as well as the total app usage time and step counts accumulated today. The screen usage card further divides the accumulated screen use time into the three time slots: morning (12am to 8am), daylight (8am to 4pm), and night (4pm to 12am), and the three most frequently used apps of each slots.

The bottom of the main page (i.e., diary region) shows the button triggering diary reporting service (described later) before filling the diary or side-by-side comparison view of patients' self reports and parents' observations after filling the diary. Once completing the diary, the adolescents (or parents) can compare their responses to questions in daily with the counterparts responded in parents' (or adolescents') diary to identify inconsistencies among them. After the adolescents and parents perceive these differences appearing in the diary region, they can choose a better time to communicate with each other over these differences.

2.1.2 Usage detail page. Figure 1(b) shows a screenshot of the usage detail page, which plots the bar chart and lists three most frequently used apps in each time slot. The top one visualizes the amount of app usage time accumulated in the three time slots, so that adolescents (or parents) can easily identify the time slot that the adolescents spend the most time on using apps. However, the adolescents might spend time on apps for academic as well as recreational purposes. So, at the bottom, the detail page lists the three most frequently used app categories [1] and the corresponding usage time. Through reviewing the time spent on using different categories of apps, both adolescents and parents can avoid mistaking the productive technology use behaviors (e.g., time using Education-type apps) with unproductive ones (e.g., time using Game-type apps).

2.2 Diary Reporting Service

Leveraging an existing online survey tool [2], we design a diary reporting service acquiring self-reports from adolescents and observational data from their parents through a diary, which contain a parallel set of probing questions. The adolescents or parents can

fill out the diary by using this app. After the adolescents or parents finish the diary each day, the phone app can automatically download the responses towards these questions to visualize in the main page through the APIs provided by the survey tool [2].

The diary contains the questions asking adolescents to self-report (or the parents to observe) adolescent's physical conditions (e.g., sleep quality, level of physical activity, or having any illness symptoms) and mental status (e.g., their moods or overall mental status), and what contributing factors (e.g., physical conditions, mental status, or family communication) made the adolescents spend more (or less) time on using the phones. For both adolescents and parents, the questions asked if they used the numbers collected or inferred from the Fitbit data (i.e., the technology sensing part), the interaction/communication between adolescents and their parents (i.e., the family supporting part), and how technology and family supporting parts raised their awareness, affected their intention, or improved their technology abuse.

2.3 Backend Server

The server compiles technology use data (app name and the associated usage time and categories) as well as in situ self-reports/observations through filling the diary. We also included a web monitoring module that correlates technology use data, self-reports of adolescents, and parent's observations. This makes it easier for treatment team to figure out how to introduce the interventions to the adolescents and parents.

3 CONCLUSION & FUTURE WORK

We will continue refining the design of TechLifeProbe and will conduct a technology probe study by recruiting adolescents with technology abuse and parents to use this system in their daily lives.

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